# 1.6 Meeting

本次会议集中讨论的**核心问题是如何利用Multi-source Data学习**:

一个中心决策者，**如何同时利用多个不同的local data set**，通过knowledge sharing或regularization，达到优化的最佳效果。

以下是对不同选题的讨论，接下来会按照不同方向进行**文献综述**。

# Joint Estimator

Joint estimator**利用global regression的系数进行正则化Regularization**：

**Joint Estimator (Regularized Regression)**

2 Joint Estimator

We adapt the joint estimator from Bastani (2021) on the global and local data as follows:

- **Step 1: Global Estimator** We run an OLS regression on the global data set $\bigcup_{j\in[J]} S^{(j)}$ and obtain a *global estimator* $\hat{\beta}_{\text{global}}$ defined as

$$\hat{\beta}_{\text{global}} = \arg\min_{\beta} \left\{ \sum_{j\in[J]} \frac{1}{N_j} \left\| \mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta \right\|_2^2 \right\}. \tag{3}$$

- **Step 2: Joint Estimator** For each data set $S^{(j)}$, we obtain a joint estimator $\hat{\beta}^{(j)}_{\text{joint}}$ via a LASSO regression. The regularization term is based on the deviation between the estimated coefficient $\hat{\beta}^{(j)}_{\text{joint}}$ and the global estimator $\hat{\beta}_{\text{global}}$. The joint estimator is a **LASSO estimator of Local Data debiased by Global Estimator**.

$$\hat{\beta}^{(j)}_{\text{joint}} = \arg\min_{\beta} \left\{ \frac{1}{N_j} \left\| \mathbf{D}^{(j)} - \mathbf{X}^{(j)\top}\beta \right\|_2^2 + \lambda^{(j)} \left\| \beta - \hat{\beta}_{\text{global}} \right\|_1 \right\}. \tag{4}$$

  Here, $\lambda^{(j)} > 0$ is a **regularization parameter** that controls the influence of the global estimator $\hat{\beta}_{\text{global}}$ in the joint model. The **$\ell_1$-norm penalty exploits the sparsity structure** of the bias term $\xi^{(j)}$.

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这里的原理是$\beta, \beta_{\text{global}}$相近，也就是利用**数据/模型相似性**进行迁移；

根据Berstimas and Kallus (2020)的方法，通过**数据的远近**进行加总：

$$\hat{y}\left(x^0\right) = \sum_{i=1}^{n} w\left(x^i, x^0\right) y^i,$$

这里**权重**$(x^i, x^0)$根据feature point和$x^0$的距离确定的，常见的local ML method（即采用nonparametric weighted neighborhood的)方法有：

k-nearest neighbor，classification and regression trees (CART) 和kernel estimation，Nadaraya-Watson (NW) kernel estimation, random forest.

按这个思路，可以**将相似的店进行cluster**，再用相同的$\beta_{\text{cluster}}$进行调优

# 1. Mixture DRO

该方法是对Federate Learning中**Mixed Model**的应用，迁移到DRO部分：

## Mixture DRO Approach

4 Method 2: Mixture DRO Approach

We borrow ideas from the conditional group DRO framework and feaderate learning (Marfoq et al., 2021; Wang et al., 2025b), to define **the local distribution as mixtures from the predicted empirical distributions**.

We can construct an ambiguity set $d_j$ conditioning on the observed covariate $\mathbf{X}$:

$$\mathcal{F}_j(\mathbf{X},\theta) = \left\{ \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \in \mathcal{P}(\mathbb{R}_+) \;\middle|\; \begin{array}{l} d_j \sim \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} \\ \mathbb{Q}^{(j)}_{\boldsymbol{d}|\mathbf{X}} = \sum_{l\in[J]} q^{(j)}_l \hat{\mathbb{P}}(\hat{\boldsymbol{\beta}}^{(l)}_{\text{joint}}, \boldsymbol{x}_j, l) \\ \boldsymbol{q}^{(j)} \in \mathcal{H}(\hat{\boldsymbol{q}}^{(j)}, \theta) \end{array} \right\}$$

where the uncertainty set $\mathcal{H}(\hat{\boldsymbol{q}},\theta) \subseteq \Delta^J$ is defined as:

$$\mathcal{H}(\hat{\boldsymbol{q}},\theta) = \left\{ \boldsymbol{q} \in \mathbb{R}^J \;\middle|\; \begin{array}{l} \mathbf{1}^\top \boldsymbol{q} = 1 \\ \boldsymbol{q} \geq \mathbf{0} \\ \|\boldsymbol{q} - \hat{\boldsymbol{q}}\| \leq \theta \end{array} \right\}$$

- **Mixture weight: $\boldsymbol{q}$ reside in the $\theta$-neighborhood of $\hat{\boldsymbol{q}}$** with respect to the general norm.
- **Prior knowledge: $\hat{\boldsymbol{q}}$ is a pre-specified weight** obtained from training.
- **Robustness budget:** $\theta \geq 0$ controls the size of $\mathcal{H}(\hat{\boldsymbol{q}},\theta)$.

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如何确定权重 **weight**，仍然按照不同分布的相似性进行mixing：

- **Kernel Method**: 参考*From Predictive to Prescriptive Analytics*， 张莲民的*Newsvendor Under Stochastic Dominance*，本质上都是根据数据的相似性进行加权，**权重由Kernel Method确定**。
- **拓展到按时间划分的分布**：本文注重不同retailer的分布需求，实际上假如同一个运营商有12个月的需求数据，最终需求预测也可以用加权；

$$\mathbb{P}_1, \mathbb{P}_2, \ldots, \mathbb{P}_{12}$$

- **同时考虑时间和空间**：即考虑时间$t$和retailer $j$的划分，再通过某种权重聚合，复杂度升高

$$\mathbb{Q}_{\boldsymbol{d}|\mathbf{X}}^{(j,t)} = \sum_{l\in[J\times T]} \boldsymbol{q}_l^{(j,t)} \hat{\mathbb{P}}$$

- **实际上是Regression+Kernel的聚合**：先用Regression确定各local residual distribution，再根据原有数据分布的相似性进行聚合。

  即比Bertsimas多了一步Regression的过程，再聚合，最后考虑权重的uncertainty set；增加一些不确定性。

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如果考虑**Mixed Regression**，则在Regression阶段就要确定权重，那么**对应的loss function会更加复杂**，类似于

- **Linear combination of loss function**: 最终loss function是原来loss function和

$$\hat{h}_{loc,i}^* = \arg\min_{h\in\mathcal{H}} \hat{\mathcal{L}}_{\mathcal{D}_i}(\alpha_i h + (1-\alpha_i)\bar{h}^*),$$

  即local model for $j$-th client is a **convex combination of estimators**.

# Regularization

将Regularization扩展到Optimize部分：

## 2. Cost function Regularization- Fairness

$$\min_{f\in\mathcal{F}} \mathbb{E}_{Z\sim\mathbb{P}_n}[f(Z)] + \rho|\mathcal{V}_{\mathbb{P}_{global}}(f) - \mathcal{V}_{\mathbb{P}_n}(f)|$$

这种通过Variation loss进行Regularize的解释比较模糊，不好定义。

不如用Cost difference进行Regularization：

$$\min_{z_j} \mathbb{E}_{d\sim\mathbb{P}_j}[C(z_j,d) + \lambda|C(z_j,d) - \hat{C}_{\text{global}}(z_j)|$$

这相当于**Fairness**的概念，加上一个Lagrange Multiplier，控制任意两个估计的cost之间不会偏差太多。在这里，是控制local cost和global cost不要相差太多。

# 3. Entropic Regularization-模糊集

考虑采用Entropic Regularize的Sinkhorn Distance:

$$\mathcal{W}_\epsilon(\mathbb{P},\mathbb{Q}) = \inf_{\gamma\in\Gamma(\mathbb{P},\mathbb{Q})} \left\{\mathbb{E}_{(x,y)\sim\gamma}[c(x,y)] + \epsilon H(\gamma \mid \mu\otimes\nu)\right\},$$

但原问题引入Entropy Regularization是为了使整个分布更平滑，而不是为了本文的$\mathbb{P}_{\text{local}}$和$\mathbb{P}_{\text{global}}$g更近，**要查清楚Entropic的作用**；可以扩展到Phi-divergence，同样只针对连续分布。

**Using global estimation as a regularization reference**: We can extend this idea, to choose reference measures $\mu$ as residual distribution from global estimator, i.e., $\hat{\mathbb{P}}_{\text{global}}$:

$$H(\gamma \mid \hat{\mathbb{P}}_{\text{global}}\otimes\nu) = \mathbb{E}_{(x,y)\sim\gamma}\left[\log\left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\hat{\mathbb{P}}_{\text{global}}\mathrm{d}\nu(y)}\right)\right],$$

where $\hat{\mathbb{P}}_{\text{global}} \ll \mu$.

可以考虑只采用Wasserstein距离

- **只采用Wasserstein距离**：既然Wasserstein距离也可以限制分布的距离，可以考虑加入global distribution的Regularization;

$$f(\boldsymbol{x},\boldsymbol{z}) - \tau \le \kappa(\Delta(\mathbb{P},\hat{\mathbb{P}}_{local}) - \Gamma)^+ + \hat{\kappa}(\Delta(\mathbb{P},\hat{\mathbb{P}}_{global}) - \hat{\Gamma})^+ \quad \forall \mathbb{P}\in\mathcal{P}_0(\mathcal{Z}) \quad (3)$$

- 或者定义一个Regularized Wasserstein距离：Regularization考虑实现分布$\mathbb{P}$和global distribution的距离；

$$\mathcal{W}_\epsilon(\hat{\mathbb{P}}_{\text{local}},\mathbb{P}) = \inf_{\gamma\in\Gamma(\hat{\mathbb{P}}_{\text{local}},\mathbb{P})} \left\{\mathbb{E}_{(x,y)\sim\gamma}[\|x-y\|^2] + \epsilon\mathbb{E}_{(x,z)\sim\Gamma(\mathbb{P},\hat{\mathbb{P}}_{\text{global}})}\left[\left(\|x-z\|^2\right)\right]\right\}$$

### Model Misspecification

既然都是要扩大Wasserstein模糊集，则可以通过**Model Misspecification**的原理，在原来的模糊集上再添加一族分布用来校正。

- 参考：